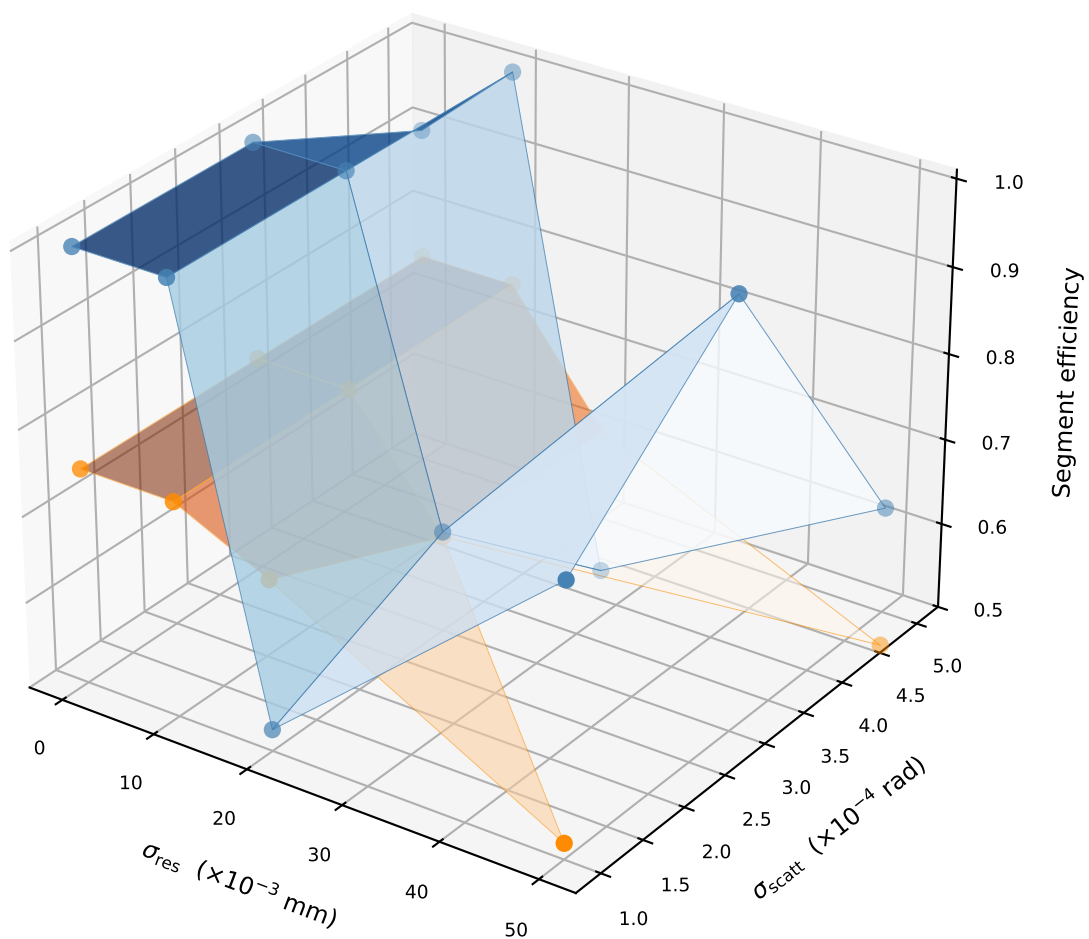
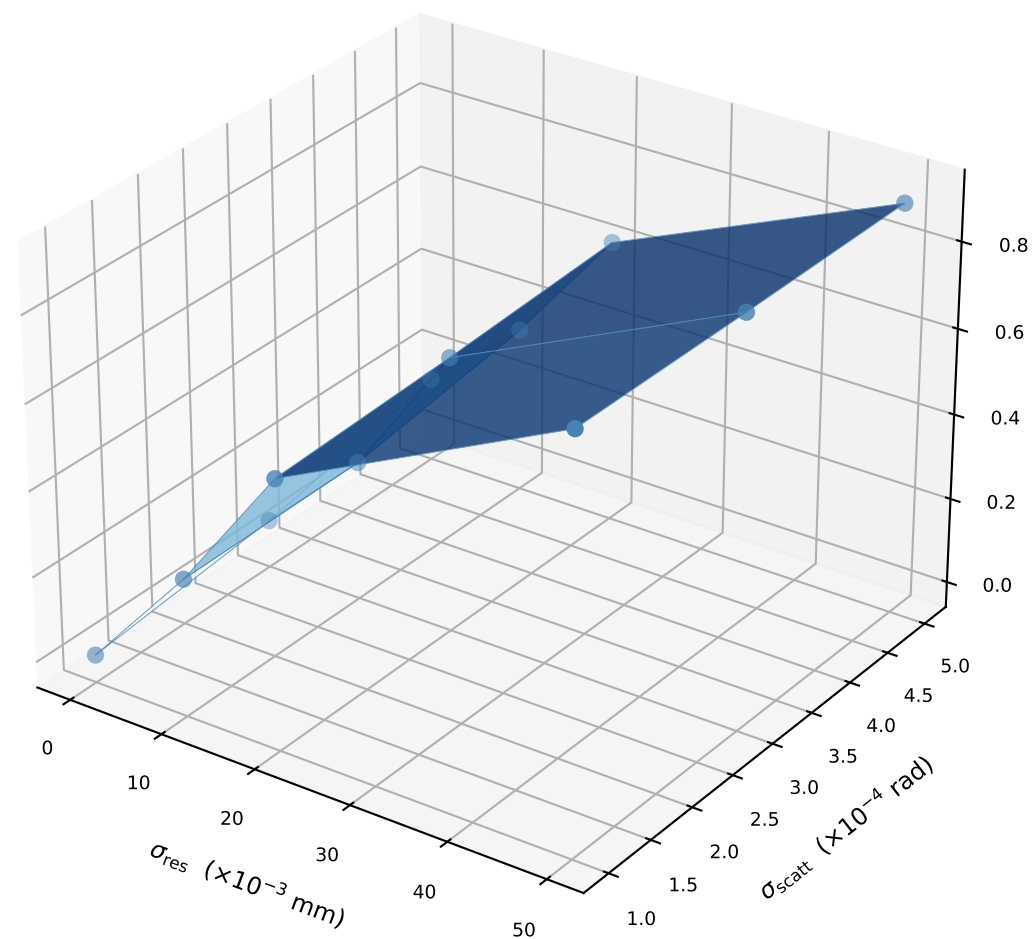


Metrics over ($\sigma_{\text{res}}, \sigma_{\text{scatt}}$) phase space — $T = 200$ tracks (reps/cell: 5-5)

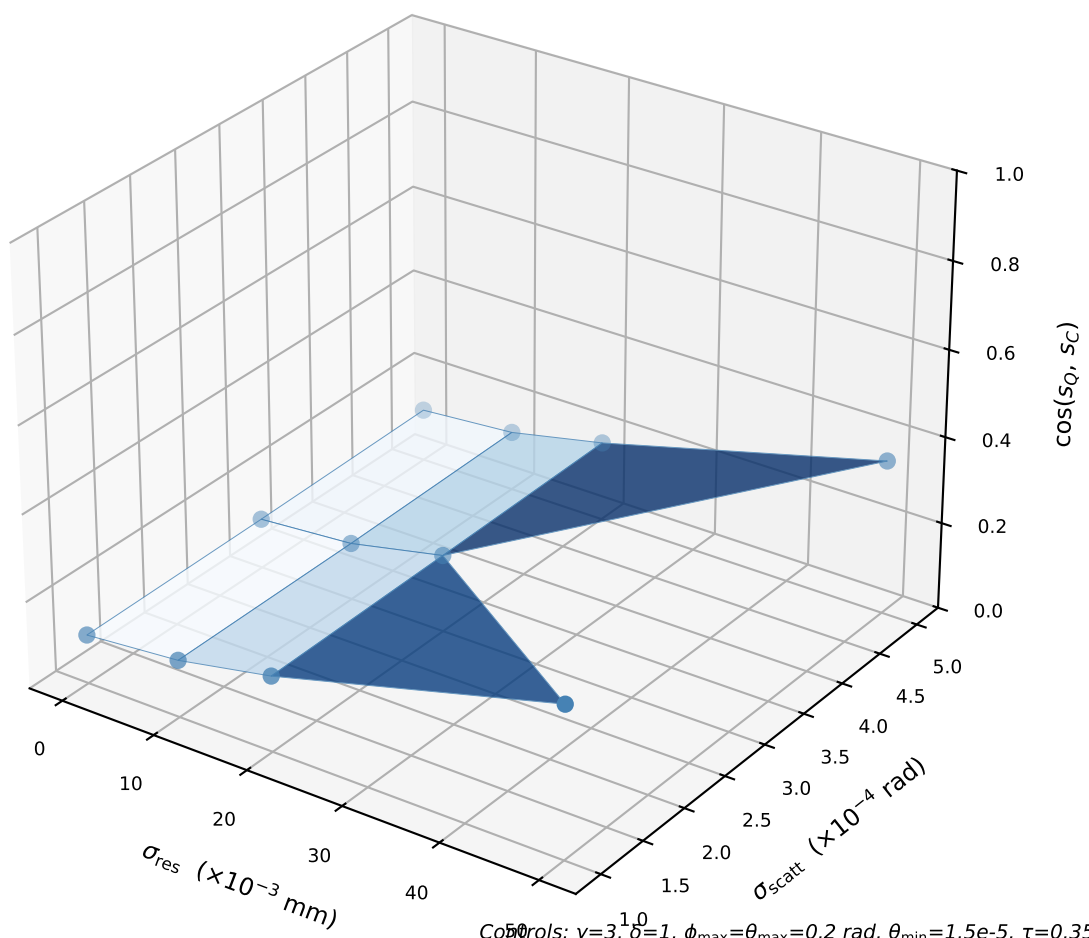
Segment efficiency (Blue=C, Orange=Q)



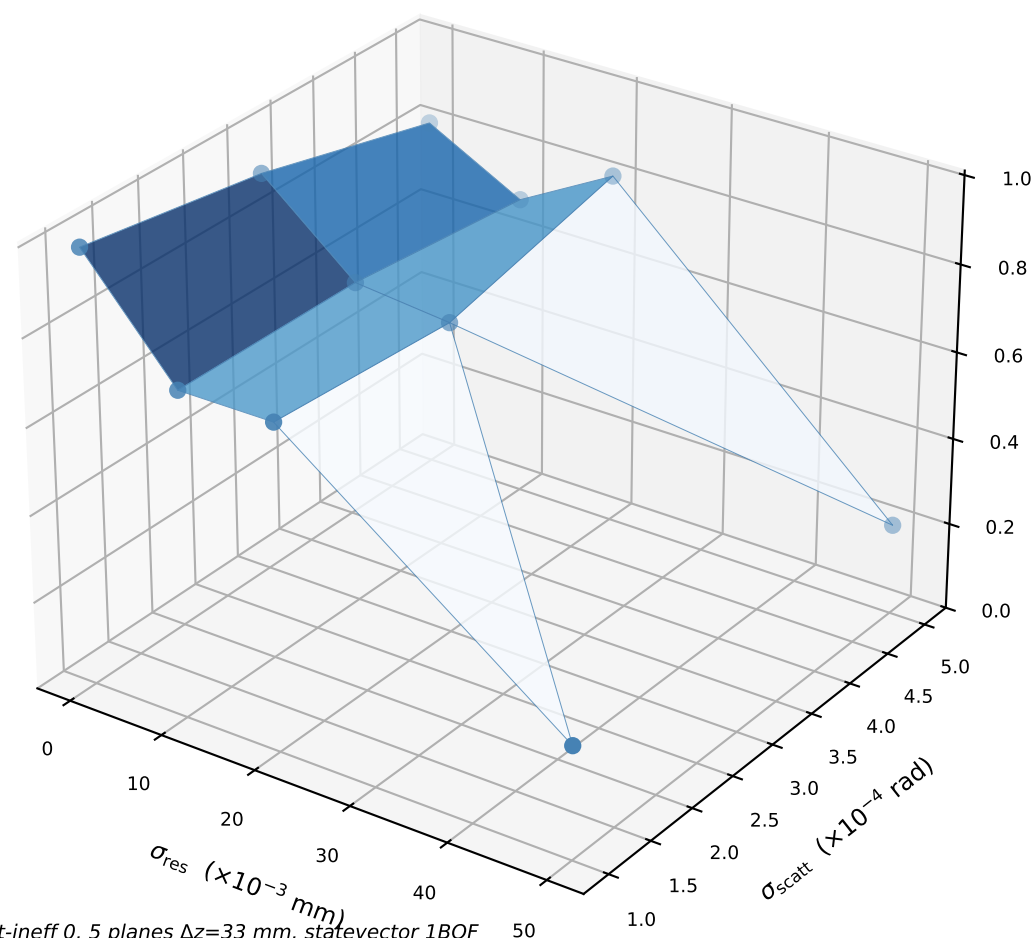
Angle false rate (Classical)



$\cos(s_Q, s_C)$ (Classical)



Quantum purity (Classical)



Controls: $\gamma=3, \delta=1, \phi_{\text{max}}=\theta_{\text{max}}=0.2 \text{ rad}, \theta_{\text{min}}=1.5\text{e-}5, \tau=0.35, \epsilon=\text{calc.}(s=3), \text{conv OFF}, \text{hit-in-eff } 0, 5 \text{ planes } \Delta z=33 \text{ mm}, \text{statevector 1BQF}$
 Fixed here: $T=200$ fixed; Blue=Classical, Orange=Quantum
 Swept: $\sigma_{\text{res}} \times \sigma_{\text{scatt}}$ (full grid)